Dr. Sagar Deep Deb

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Education

Indian Institute of Technology Patna — Patna, Bihar, India
PhD in Electrical Engineering; — *July 2019 – May 2024*

National Institute of Technology Silchar — Silchar, Assam, India
MTech in Communication and Signal Processing; ***GPA: 8.82/10*** — *June 2017 – June 2019*

North Eastern Regional Institute of Science and Technology — Nirjuli, Arunachal Pradesh, India
BTech in Electronics and Communication Engineering; ***GPA: 3.60/5*** — *June 2012 – May 2016*

Experience

University of Petroleum and Energy Studies, Dehradun — Dehradun, Uttarakhand, India
Assistant Professor, School of Computer Sciences — *September 2025 – Present*

Indian Institute of Technology Bombay — Mumbai, Maharashtra, India
Institute Post Doctoral Fellow, Department of Biosciences and Bioengineering — *June 2024 – September 2025*

Indian Institute of Technology Madras — Gandhinagar, Gujarat, India
Senior Project Officer, Department of Applied Mechanics and Biomedical Engineering — *Feb 2024 – May 2024*

Skills

Programming: Python, Keras, MATLAB, R

Languages: Bengali (Native), English (Professional), Hindi (Professional)

Publications (Journals)

- Arjun Abhishek, **Sagar Deep Deb**, Rajib Kumar Jha, Ruchi Sinha, & Kamlesh Jha (2025). Ensemble learning using Gompertz function for leukemia classification. *Biomedical Signal Processing and Control, 100, 106925.*
- **Sagar Deep Deb**, Md.Aqhlakur Rahman, and Rajib Kumar Jha. "Breast Cancer diagnosis using Modified Xception and Stacked Generalization Ensemble Classifier." *Research on Biomedical Engineering (2023): 1-14.*
- **Sagar Deep Deb**, and Rajib Kumar Jha. "Segmentation of Pectoral Muscle from Mammograms using U-Net having Densely Connected Convolutional Layers." *Multimedia Tools and Applications (2023): 1-22.*
- **Sagar Deep Deb**, and Rajib Kumar Jha. "Breast Ultra Sound Image Classification using fuzzy-rank-based ensemble network." *Biomedical Signal Processing and Control 85 (2023): 104871.*
- **Sagar Deep Deb**, Rajib Kumar Jha, Rajnish Kumar, Prem S. Tripathi, Yash Talera, and Manish Kumar. "CoVSeverity-Net: an efficient deep learning model for COVID-19 severity estimation from Chest X-Ray images." *Research on Biomedical Engineering (2023): 1-14.*
- **Sagar Deep Deb**, and Rajib Kumar Jha. "Modified Double U-Net Architecture for Medical Image Segmentation." *IEEE Transactions on Radiation and Plasma Medical Sciences (2022).*
- **Sagar Deep Deb**, Rajib Kumar Jha, Kamlesh Jha, and Prem S. Tripathi. "A multi model ensemble based deep convolution neural network structure for detection of COVID19." *Biomedical Signal Processing and Control 71 (2022): 103126.*

Publications (Conferences)

- **Sagar Deep Deb**, Rajib Kumar Jha, and Sudhir Kumar. "ConvPlant-Net: A Convolutional Neural Network Based Architecture for Leaf Disease Detection in Smart Agriculture." In 2023 National Conference on Communications (NCC), pp. 1-5. IEEE, 2023.
- **Sagar Deep Deb**, Arjun Abhishek, and Rajib Kumar Jha. "2-Stage Convolutional Neural Network for Breast Cancer Detection from Ultrasound Images." In 2023 National Conference on Communications (NCC), pp. 1-5. IEEE, 2023.
- Arjun Abhishek, **Sagar Deep Deb**, and Rajib Kumar Jha. "Effective WBC Segmentation using Hybrid Loss." In 2023 National Conference on Communications (NCC), pp. 1-5. IEEE, 2023.

- **Sagar Deep Deb**, Rajib Kumar Jha, Yash Talera, Prem S. Tripathy, Simran Agrawal and Kamlesh Jha, "Survival prediction of COVID-19 patients using multi-modal dataset," 2022 IEEE 19th India Council International Conference (INDICON), pp. 1-6. IEEE, 2022
- **Sagar Deep Deb**, and Rajib Kumar Jha. "Covid-19 detection from chest x-ray images using ensemble of cnn models." In 2020 International Conference on Power, Instrumentation, Control and Computing (PICC), pp. 1-5. IEEE, 2020.
- **Sagar Deep Deb**, Md Akhlaqur Rahman, and Rajib Kumar Jha. "Breast cancer detection and classification using global pooling." In 2020 11th International Conference on Computing, Communication and Networking Technologies (ICCCNT), pp. 1-5. IEEE, 2020.
- **Sagar Deep Deb**, Chandraiit Choudhury, Manish Sharma, Fazal Ahmed Talukdar, and Rabul Hussain Laskar. "Frontal facial expression recognition using parallel CNN model." In 2020 National Conference on Communications (NCC), pp. 1-5. IEEE, 2020.

Awards & Achievements

Best paper award at International Conference on Signal Processing, Computation, Signal Processing, Power and Telecommunication (IConSCET) held at National Institute of Technology Puducherry (2023)

Travel Grant for Graduate Students' Day National Conference on Communications, 2023, IIT Guwahati

Institute fellowship grant from MHRD, Govt. of India:

Relevant Coursework

Major coursework: Machine Learning, Digital Image Processing, Deep Learning, English Communications.

Minor coursework: Artificial Intelligence, Image and Video Processing.

References

Dr. Rajib Kumar Jha Associate Professor, Department of Electrical Engineering, Indian Institute of Technology Patna.
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Prof. Preetam Kumar Professor, Department of Electrical Engineering, Indian Institute of Technology Patna.
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